

DANIEL HALWELL

Northwich, Cheshire danielhalwell@gmail.com codehalwell.io github.com/CodeHalwell +447939481907

SUMMARY A results-driven Senior Scientist working at the intersection of Pharmaceutical Science and AI Engineering, with proven experience delivering AI solutions in fast-paced and highly regulated pharmaceutical environment. Expert in Python, Generative AI, Advanced ML algorithms, with hands-on experience building REST APIs and data pipelines. Combines rigorous scientific validation with agility - from troubleshooting manufacturing issues on the floor to deploying AI solutions with rapid MVP delivery and iterative improvements. Passionate about applying cutting-edge technologies across ML and AI to solve complex challenges. Thrives in collaborative, cross-functional environments where scientific rigor meets rapid innovation.

KEY SKILLS

- Proficient in **Python** and libraries like **Pandas, NumPy, Scikit-Learn, PyTorch**, and **TensorFlow**.
- Experienced in **Agentic Frameworks** such as **CrewAI, SmolAgents, Llamaindex, OpenAI Agents SDK** and **LangGraph**
- Experienced with **Generative AI** SDK's such as **OpenAI, Anthropic, OpenRouter, Google Vertex**.
- Developed REST APIs with **FastAPI** and **Flask**.
- I maintain a personal portfolio website with **TypeScript** and **Next.js**
- Familiar with Cloud platforms - **Azure, AWS and GCP**
- Experienced in the use of **Docker** for containerisation.
- Avid user of **GitHub Copilot, Gemini Code Assist**, as well as **Claude Code, Codex** and **Gemini-CLI**.
- Expertise in data visualization using **Power BI, Matplotlib, Streamlit**, and **Plotly** for stakeholder communication.

ACHIEVEMENTS

- Achieved **4th place** in a Kaggle competition (Binary Classification) in March 2025.
- **Won the \$5000** Modal Labs Choice Award for my Agentic Model Context Protocol (MCP) server during a Gradio + Huggingface hackathon in July 2025

WORK EXPERIENCE

CoDHe Labs | AI Engineer (Part-Time)

July 2025 - Present

- Providing guidance on AI implementation with a focus on generative AI, ML model development, and data analysis for various industries, especially for non-profits.
- This includes offering digital support, digital uplift strategy and managing services like Azure or M365.
- I have built an agentic proof of concept VSCode extension for a client in the healthcare sector, used for mapping of patient data.

AstraZeneca | Senior Scientist

May 2021 - Present

- **AI & Agentic Workflows:** Led the development of a RAG-based Generative AI chatbot as a laboratory assistant, cutting troubleshooting lead times by 20%. This included a multi-disciplinary team of local subject matter experts, AI engineers and front/back-end developers. I worked through the whole workflow from proof of concept to risk assessments, AI evaluation against expected output and user acceptance testing. I also led the development of low-code solutions through Microsoft Copilot Studio, working directly with Microsoft, and actively developing Agentic workflows to reduce method development by up to 80%.
- **Data Analysis & Engineering:** Engineered data pipelines using SQL (Snowflake) and Python for data analysis, data cleaning, preparing machine learning models, and deploying Streamlit applications. Successfully launched two REST APIs utilizing FastAPI, to connect critical data streams together, in preparation for GenAI applications, via tool use and MCP.
- **Data Science Initiatives:** Leveraged Python for automating tasks and developing predictive models, which resulted in a 50% reduction in method development time using Bayesian optimisation. Developed predictive models to increase efficiency of chemical testing by utilising in-silico modelling.
- **Building Dashboards:** I developed a suite of impactful dashboards to drive departmental efficiency and data-driven decision-making. These tools provide a consolidated view of support tickets, instrument utilization and self-serve portal for nitrite data that has significantly streamlined analytical workflows.
- **Technical Leadership & Mentorship:** Played an active role in the internal Coding Network by sharing knowledge and mentoring peers in data science and programming and currently on the programming expert panel.

- **Technical Support & Data Analysis:** Collaborated with scientist, engineers and microbiologists to troubleshoot issues, from chemicals to bacterial/mould contamination within manufacturing environments. Developed critical analytical methods on-the-fly to solve urgent production issues.
- **Extractables and Leachables:** I was a subject matter expert on Extractable and Leachable studies, leading studies both internally and externally for multiple products, ensuring compliance with regulatory bodies.
- **Problem-Solving:** Led cross-functional team using Practical Problem Solving (PPS) to resolve a critical impurity issue, generating cost savings of £100,000 for the company
- **Statistical Analysis:** Performed routine statistical analysis to support business decisions and regulatory submissions.

Sanofi | Quality & Senior Quality Analyst

October 2012 - September 2016

- **Routine Analysis:** Responsible for routine testing of drug substance and drug products in a QC environment. The testing was performed in a cGXP environment, ensuring compliance to regulatory standards.
- **Investigations:** As an experienced member of the team, I led out-of-trend and out-of-specification investigations across multiple disciplines such as engineering, manufacturing and quality.
- **Leadership:** Deputised for line management during absences, including managing team workload and planning.

Mars Petcare | Research Scientist

May 2009 - August 2011

- **Summer Placement - Research Assistant at Mars Petcare (Jun 2011 - Aug 2011)**
 - Explored the link between flavor perception and compound concentration in the mouth.
 - Introduced the sensory analysis software FIZZ, including set-up and document writing.
 - Utilized analytical techniques such as direct injection mass spectrometry with a triple quad mass analyzer (MS/MS).
- **Industrial Placement - R&D Scientist at Mars Petcare (Aug 2009 - September 2010)**
 - Worked in the analytical lab using equipment like LC-MS, GC-MS, and GC-FID.
 - Conducted organic chemistry research on chemical interactions in food cooking to enhance flavor compound formation.
 - Assisted in pilot-scale production of new food formulations and conducted sensory trials and result interpretation.

Project Trust | Volunteer Teacher

August 2006- June 2007

- Taught Mathematics, Science, and Physical Education to secondary school students (11-16 years old) in Guyana, South America.
- Focused on developing and implementing engaging lesson plans that not only met educational standards but also cultivated a genuine interest in learning for my students.

EDUCATION

Masters of Chemistry	2007 – 2012
Loughborough University	
• Achieved a 2:1 with a diploma in industrial studies	
Professional Certificates	2023 – 2025
• Docker Foundations Professional Certificate • Microsoft Python Development Certificate • IBM AI Developer Professional Certificate • Google Data Analytics Professional Certificate • DeepLearning.AI TensorFlow Developer Specialization • DataCamp - SQL Associate certificate • DataCamp - Data Scientist Associate certificate • IBM RAG and Agentic AI Professional Certificate	- LinkedIn - Coursera - Coursera - Coursera - Coursera - DataCamp - DataCamp - Coursera
Professional Certificates (In Progress)	2025 - Present
• IBM Generative AI Engineering • IBM DevOps and Software Engineering	- Coursera - Coursera

- INTERESTS
- **Passionate Coder & Lifelong Learner:** Constantly exploring new concepts in technology and beyond.
 - **Interdisciplinary Interests:** Fascinated by and enjoy researching about the sciences, economics, neuroscience, and psychology.
 - **Practical Project Builder:** In my spare time I like working on projects creating chatbots, automations, and other mini-projects using platforms like Replit, n8n and pure Python..
 - **Keen Sports Enthusiast:** An avid follower of Football, Rugby, and Formula 1.